

Chenyu (Fred) Zhang

New York, NY | (667) 213-8216 | zhangchenyu555@gmail.com | fredzcy.github.io

WORK EXPERIENCE

Data Scientist | Morgan Stanley

New York, NY | Feb 2023 – Present

- Built and deployed a **production RAG knowledge assistant** for **700+ users**, indexing thousands of metadata-enriched chunks from enterprise documents with structure-aware parsing, parent-child chunking, **Redis vector search**, hybrid retrieval, reranking, and prompt grounding to deliver source-backed answers with low-latency responses.
- Developed **reusable evaluation and governance frameworks** across GenAI applications, including golden-query regression tests, AI-assisted response scoring, human review workflows, logging, prompt/version tracking, and failure-mode analysis to improve reliability in regulated enterprise deployments.
- Built **agentic GenAI workflows** using **OpenAI Agents SDK** for institutional securities data access and process automation, combining semantic schema linking, SQL validation guardrails, controlled database execution, **MCP-based tool invocation**, specialized creator/reviewer agents, and critique-and-repair loops to generate validated SQL outputs, process flowcharts, and walkthrough narratives; estimated to **save up to 80 hours per workflow**.
- Developed a **market news intelligence system** processing **6,000+ daily financial news items**, combining semantic embeddings, **BM25 keyword retrieval**, and **Reciprocal Rank Fusion (RRF)** ranking to surface market events and commentary aligned with user-defined research objectives, sectors, and risk themes.
- Built an **LLM-powered sentence-level classification pipeline** over **32,000+ multilingual client interaction records**, using language detection, translation, sentence embeddings, threshold calibration, and interpretable review outputs to detect **10+ compliance violations** and improve review consistency.
- Designed a **hybrid NLP issue-intelligence pipeline** over **9,000+ issue records**, combining keyword matching, zero-shot classification, and **topic modeling** to detect emerging risk themes, support scalable analysis and executive-level reporting.
- Developed **production Python services and data pipelines** for AI applications, integrating **Snowflake, PostgreSQL, Redis, FastAPI, Docker/Kubernetes**, CI/CD workflows, and automated testing to support scalable enterprise deployment.
- Drove **AI adoption** through hands-on training, demos, technical documentation, and **8 technical presentations** on NLP, Explainable AI, and LLMs, supporting rollout across **10+ teams** and accelerating trust in AI-assisted workflows.

Data Science Summer Intern | Morgan Stanley

New York, NY | Jun 2022 – Aug 2022

- Developed a **neural-network-based dashboard recommendation model** for **700+ users** across **100+ dashboards**, combining user interaction signals and dashboard metadata to personalize discovery of internal analytics tools.
- Implemented a **multimodal deep learning classification model** in **TensorFlow/Keras** over **23,000+ enterprise issue records**, integrating unstructured text, structured business metadata, and data augmentation to improve rare-category detection, achieving **93% recall** in quarterly tagging workflows.

Research Assistant | Johns Hopkins University / IEEE

Baltimore, MD | Mar 2022 – Oct 2022

- Built an **AWS-based large-scale data normalization pipeline** over **700M+ IEEE publication records**, integrating external APIs, BERT embeddings, cosine similarity, and hierarchical clustering for institutional affiliation disambiguation.
- Improved processing efficiency by **~90% through multiprocessing**, achieving **89% disambiguation accuracy** across noisy institutional affiliation records.

EDUCATION

Johns Hopkins University | Baltimore, MD

M.S.E. in Data Science | GPA: 3.73/4.0 | Dec 2022

Xiamen University | Sepang, Malaysia

B.E. in Computer Science and Technology (Honors) | GPA: 3.89/4.0 (*Ranked 1/61*) | Aug 2021

SKILLS

Programming Languages: Python, SQL, R, C++, JavaScript, HTML

GenAI & Agentic AI: RAG (Graph RAG), LangChain, LangGraph, OpenAI Agents SDK, FastMCP, LLM evaluation

ML & Data: NumPy, Pandas, Scikit-learn, TensorFlow/Keras, PyTorch

Application & Infrastructure: FastAPI, Flask, REST APIs, AWS, Docker, Kubernetes, Kafka, CI/CD (Jenkins), Git, Linux, Grafana

Databases & Platforms: PostgreSQL, DB2, Snowflake, Redis, Neo4j, Spark, Databricks

Certifications & Languages: CFA Level I; English (Fluent), Mandarin (Native)